

An energy-aware broadcast scheme for directed diffusion in wireless sensor network

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Abstract: Broadcast is a common communication means in mobile ad hoc network or wireless sensor network. In directed diffusion protocol, broadcast is used to disseminate interest, discover routing path and repair the routing path. However, straightforward broadcast scheme (such as flooding) may result in serious broadcast storm problem, especially with respect to large densely deployed wireless sensor network. This paper compares the performances of several broadcast schemes combining with directed diffusion and proposes an energy-aware broadcast scheme adapting to directed diffusion. This scheme combines the advantages of distance-based and counter-based broadcast schemes and considers the remainder energy of sensor node. Simulation results show that the proposed scheme has the following advantages: (1) simple, efficient and scalable; (2) more energy-balancing than other schemes; (3) longer network lifetime than other schemes.

Key words: wireless sensor network; directed diffusion; broadcast; energy-balancing

1. Introduction

With the rapid development of micro-electro-mechanical systems (MEMS) technology, wireless communications and digital electronics, wireless sensor network consisting of large number of low-cost, small-size and multifunctional sensor nodes is walking out from research papers and becoming reality. Its appearance can revolutionize information gathering and processing in many

situations. For example, wireless sensor network deploys in inhospitable physical environments such as disaster area or aircraft interiors can gather and process information of the whole target region which is an impossible task for a sensor node.

Compared with mobile ad hoc network (MANET), wireless sensor network (WSN) concentrates on gathering and transferring data. Due to the high redundancy of nodes in WSN, any individual node is not particular important, it might not even have a unique address. An appropriate programming model for such a network is publish/subscribe, with data-centric routing as the supporting communication layer. Directed diffusion (DD)^[1] is a prominent data dissemination paradigm for WSN. Data generated by sensor nodes is named by attribute-value pairs. A node requests data by sending interests for named data. Data matching the interest is then “drawn” down towards that node. DD is a data-centric routing protocol which selects routing path based on application data.

In DD, interest dissemination is accomplished by performing network-wide broadcasts. Routing path is discovered during the process. In order to repair the degraded routing path, interest is periodically broadcasted over the network from the sink. So, the overall performance of the protocol can be strongly influenced by the efficiency of the elementary broadcast operation.

Broadcast is a common communication means in MANET or WSN to resolve many issues. In some circumstances, it is even the only way of data dissemination and routing discovery. The simplest

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broadcast scheme is flooding where every node in the network forwards each new message to all of its neighbors. However, broadcast may lead to severe broadcast storm problem^[2] if it is done by simple flooding in a contention-based wireless channel with carrier sense multiple access (CSMA), with no collision detection (CD) capability. The broadcast storm problem includes serious redundant rebroadcast, contention for channel and packet collision. Ni, et al.^[2] show that a rebroadcast can provide only 61% additional coverage (AC) at most and only 41% AC in average through analysis. Assume there are n nodes hearing the same broadcast message. The probability of n nodes experiencing contention increases quickly over 0.8 as $n \geq 6$. So the broadcast storm problem deserves serious studies. In DD, especially, this problem can result in longer interest dissemination time, larger energy dissipation of sensor node and shorter network lifetime. So, an energy-efficient broadcast scheme needs to be proposed to solve this problem.

Two directions to alleviate this problem are to reduce the possibility of redundant rebroadcasts and differentiate the timing of rebroadcasts. Following these directions, Ni, et al.^[2] develop several broadcast schemes, called probability-based, counter-based, distance-based, location-based, and cluster-based schemes, to resolve this problem. Williams, et al.^[3] add a new family, namely, neighbor-knowledge-based scheme. The above six broadcast schemes are the basic methods to resolve the broadcast storm problem.

Zhu, et al.^[4] proposed a border-aware broadcast scheme combining the advantages of distance-based and counter-based methods. Kim, et al.^[5] propose a probabilistic broadcasting based on coverage area and neighbor confirmation which combine the advantages of distance-based and probability-based methods.

Although many broadcast schemes have been proposed, how to incorporate the schemes with concrete routing protocol is not considered. This paper compares and analyzes the performances of several

broadcast schemes incorporating with DD protocol, then proposes an energy-aware broadcast scheme which combines the advantages of distance-based and counter-based methods and considers the remainder energy of sensor nodes. The scheme trades off the efficiency of broadcast scheme with energy-balancing among nodes which resolves the hot spots problem and prolongs the lifetime of network.

The rest of this paper is organized as follows. Section 2 reviews the broadcast storm problem and existing broadcast schemes. Section 3 describes the proposed energy-aware broadcast scheme. Section 4 compares and analyzes the performances of several broadcast schemes incorporating with DD. Section 5 gives out the conclusions and future work.

2. The Broadcast Storm Problem

Assume network nodes share the same wireless channel, the radio transmitting range is same and contention-based MAC protocol-CSMA/CA (i.e. 802.11^[6]) is used. Broadcast is a communication means between one node and all of its neighbors. Its characteristics are spontaneous and unreliable. Any node can issue a broadcast message at any time. Little or no local information may be collected in advance. In 802.11 MAC, neither acknowledgement nor virtual carrier sense mechanisms will be used for broadcast packet. Nodes can identify the duplicate message by message ID.

2.1 Problem description

In this circumstance, if flooding scheme is used to broadcast message, namely, every node rebroadcasts every new broadcast message, the following consequences arise:

(1) Redundant rebroadcasts: In a network consisting of n nodes, a network-wide broadcast message will be sent for n times. If a node sends the message to its neighbors, all its neighbors have the message, the rebroadcast is completely redundant.

(2) Contention: After a node sends a broadcast message, all its neighbors will rebroadcast the message. If the radio ranges of some nodes overlay each other, they will compete for the common channel. If contention happens, nodes have to do backoff operation which leads to longer transmission time.

(3) Collision: Three reasons lead to packet collision. First, backoff mechanism does not entirely differentiate the timing of transmissions of different nodes. Second, half-duplex radio transceiver does not provide the Collision Detection (CD) capability. When they are transmitting, collision is not noticed. Third, the lack of RTS/CTS dialogue mechanism leads to the increase of hidden terminal problem occurs at receivers.

The broadcast storm problem results in network congestion, information transmission delay and large waste of energy.

2.2 Existing schemes

According to reference [2] and [3], there are six basic solutions to the broadcast storm problem.

(1) Probability-based

Predetermine a probability P . On receiving a broadcast message, node will rebroadcast the message based on P . Clearly, the scheme will reduce redundancy, contention and collision when P is less than 1. A small random delay is inserted before rebroadcasting the message to differentiate the timing of rebroadcasting.

(2) Counter-based

In reference [2], simulation results show the expected additional coverage (EAC) is below 0.05% after the node has heard the message for 4 times. Specially, a counter c is used to keep track of the number of times the broadcast message has been received. A counter threshold C is chosen. Every node selects a random number of time slots. Before the time slots decrease to zero, if $c \geq C$, the rebroadcast is inhibited.

(3) Distance-based

This scheme is to determine whether to rebroadcast or not, which based on the distance d between the receiver and the transmitter. Ni, et al.^[2] have analyzed the relationship between the distance d and the additional coverage (AC). If d is larger, the additional coverage will be larger. The distance d can be estimated from the signal strength on which a message is received. Specially, a distance $d_{\min}$ is used to keep track of the minimum distance between the node and broadcast transmitters. A distance threshold D is chosen. Every node selects a random number of time slots. Before the time slots decrease to zero, if $d_{\min} \leq D$, the rebroadcast is inhibited.

(4) Location-based

When the locations of the broadcasting nodes are acquired, more precise estimation to the AC is possible. The approach is supported by GPS (Global Positioning System) or other localization methods. An area threshold A ($0 < A < 0.61$) is chosen. Every node selects a random number of time slots. Before the time slots decrease to zero, if $AC \leq A$, the rebroadcast is inhibited.

(5) Cluster-based

The network is divided into some clusters by using the cluster formation algorithm. When a node receives a broadcast message, if it is neither cluster header nor gateway, the rebroadcast is inhibited. The rebroadcasts of the cluster header and gateway are determined by probabilistic, counter-based, etc.

(6) Neighbor-knowledge-based

The knowledge of 1-hop^[3] or 2-hop^[7] neighbors is obtained by periodically exchanging “Hello” messages among neighbor nodes. Whether to rebroadcast or not is determined based on some graph algorithms.

(7) Some improved schemes

Zhu et al.^[4] propose a border-aware broadcast scheme. They replace the random delay with the delay related to the distance between the receiver and the transmitter. The further a receiving node is away from

the sender, the fewer slots it waits before rebroadcasting. Before the timer expires, node determines whether to rebroadcast or not, according to counter-based scheme and distances between it and senders.

Kim, et al.^[5] propose a probabilistic broadcasting based on coverage area and neighbor confirmation. They utilize coverage area of node to determine the probability in which node rebroadcasts. In order to repair the early die-out problem, the scheme adopts neighbor confirmation method. In this method, after a given amount of time t , nodes which do not rebroadcast based on the probability check if all the neighbors have received the broadcast packet. If not, they rebroadcast the packet.

3. Energy-aware broadcast scheme

This paper's goal is to design a simple, efficient broadcast scheme to improve the interest dissemination in DD. Interest dissemination by flooding makes DD more robust, because the maximum number of paths between source and sink are set up. The optimal path (e.g. lowest latency path) can be selected to reinforce. But cost of flooding is too large, a trade-off scheme between robustness and cost needs to be proposed.

Energy-balancing is an important factor that should be considered in design of WSN. If the network communication always concentrates on several paths, the energy of the nodes lying in the paths will be depleted earlier than other nodes. Although some nodes may have energy, the network might not function. In order to make the network function as long as possible, energy load should be evenly distributed among all the nodes. For example, LEACH^[8] adopts the algorithm rotating cluster head positions in each round to balance the energy load. HEED^[9] considers the remainder energy as the factor determining whether to become cluster head or not.

Based on border-aware scheme, this paper proposes an energy-aware broadcast scheme. This scheme is described in detail in the following.

3.1 Distance estimation

When a node receives a broadcast packet, it first checks the strength of the received signal which associating with power of received signal, and estimates the distance from the received power using free space wireless propagation models. The free space equation is shown in equation (1):

$$P_r = \frac{P_t \times G_t \times G_r \times \lambda^2}{(4\pi d)^2 \times L} \quad (1)$$

where P_r and P_t are the received and transmitted signal power, respectively. G_t and G_r are the antenna gains of transmitter and receiver, respectively. λ is the wavelength of electromagnetic wave. L is the system loss factor. Assume all the nodes use the same RF devices and P_t , P_r can be estimated from the signal strength indicator of the RF transceiver chip. Then, the distance d can be calculated from the formula (1).

3.2 Energy and distance aware segment delay

Random delay inserted before rebroadcast is a blind delay which does not consider direction and energy. Border-aware delay only considers the direction. This can result in serious hot spots problem. In DD, the process of interest propagation is also the process of the routing path which is discovered. When a further node has little remainder energy, it will also rebroadcast in border-aware scheme. The node will die earlier after a amount of communications and the network life will also end. The remainder energy and distance are linked to the delay. A value $k(0 < k < 1)$ is generated:

$$k = \alpha \frac{E_{remainder}}{E_{initial}} + \beta \frac{d^2}{R^2} \quad (2)$$

where α and β are weighted values associated with energy and distance, respectively, and $\alpha + \beta = 1$. $E_{remainder}$ is the remainder energy, $E_{initial}$ is the

initial energy of the node, d is the distance between the receiver and the transmitter, and R is the radius of the RF communication of the node.

However, when the number of time slots is only determined by k , the more nodes whose k are close, the more chance of rebroadcast collision. So the window of time is further divided slots into 10 segments. Each segment has fixed number of time slots. Figure 1 shows the distribution of the time slots.

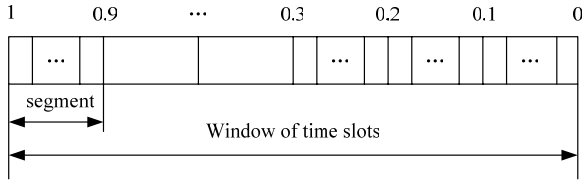


Fig. 1 Distribution of time slots

Each node keeps a rebroadcast timer. When a node receives a new broadcast message, it first calculates k , if $0.9 < k \leq 1$, the node selects a random time slots in the first segment of the window, and sets the timer to the value of the random time slots. The larger k is, the more remainder namely node has and the further from the broadcast transmitter, the smaller the number of waiting time slots is. Then the node has more chance to rebroadcast and becomes a node in the routing path.

3.3 The size and slot time of delay window calculation

The size of delay window is determined by the network density. The window size should be dynamically adjusted to different network densities. Due to the broadcast lack of acknowledge mechanism, the sender does not know whether collision occurs at the receiver or not. The slot time is enough long for a node to finish a rebroadcast so that the two consecutive broadcasts will not collide. The slot time is determined by the size of the broadcast packet and data rate of RF device.

3.4 Counter-based and distance-based rebroadcast suppression

Combine the advantages of the counter-based and distance-based schemes. Each node keeps a counter

recording the times for which the same broadcast message has been received and the minimum distance among the distances from the senders. When a node receives an old broadcast message, it checks the two variables, if counter is larger than counter threshold C or the minimum distance is less than the distance threshold D , the rebroadcast timer is canceled. Using the method, redundant rebroadcasts are reduced dramatically. The optimal values for C and D are selected by simulation.

The above four components compose the simple, efficient broadcast scheme. Energy and distance aware segment delay mechanism enhances the efficiency of the broadcast, balances the energy consumption and reduces the collision. Redundant rebroadcasts are reduced dramatically by using counter-based and distance-based rebroadcast suppression method.

4. Performance Evaluation

Counter-based scheme, distance-based scheme, border-aware scheme, and our energy-aware broadcast scheme are added to directed diffusion in ns2^[10]. In directed diffusion, there are three main algorithms: two-phase pull, push, and one-phase pull. Here the two-phase pull algorithm is changed which is implemented by gradient filter.

This section first compares and analyzes the performances of the four broadcast schemes and flooding broadcast scheme, Then compares and analyzes the performances of directed diffusion with interest dissemination using different broadcast schemes.

4.1 Performance metrics

Four performance metrics are used to evaluate the performance of DD with different interest broadcast schemes.

(1) Network Lifetime (NL): the time when the first node dies.

(2) Average Dissipated Energy (ADE): the ratio of the total dissipated energy per node in the network to

the number of distinct events seen by sinks. This metric computes the average work done by a node in delivering useful data information to the sinks.

(3) Average End-to-End Delay of data packets (AED): all possible delays during data dissemination from the source to the sink, caused by buffering, queuing at the interface queue, retransmission and back off delays at the MAC, and the propagation and transfer times.

(4) Distinct Event Delivery Ratio (DEDR): the ratio of the number of received distinct events to the number originally sent by sources.

These metrics can be studied as a function of sensor network size and density.

4.2 Simulation environment

In order to study the performance of different broadcast schemes and DD as a function of network size, five sensor fields of different sizes are generated, ranging from 50 to 250 nodes in increments of 50 nodes. The average network density is kept about 10, namely about 10 nodes within the RF communication range of a node. This 50-node sensor field is generated by randomly placing the nodes in a 156×156m square. Other sensor fields are generated by scaling the square.

However, when the impact of network density on the schemes is studied, six sensor fields of different square area are generated, ranging from 40×40m to 240×240m in increments of 40×40m, keeping the number of sensor nodes 100.

In each sensor field, there is a pair of source and sink. Source node is randomly selected in the up-left corner of the field, and sink node is selected in the down-right corner of the field. Each node has a RF communication range of 40m. All the nodes are assumed stationary.

The parameters used in this simulation are shown in Table 1 and Table 2.

Table 1 Application parameter configuration

Interest interval	30s
ata interval	5s
Exploratory data interval	60s
Data packet length	148byte
Interest packet length	120byte
Simulation duration	200s
Reinforcement mode	Lowest delay path
Slot time	2ms

Table 2 802.11MAC parameter configuration

Data rate	1Mbps
Idle power dissipation	0.035W
Receive power dissipation	0.395W
Transmit power dissipation	0.66W
Node initial energy	20J

4.3 Simulation results

The first experiment compares energy-aware broadcast scheme with flooding scheme, counter-based scheme, distance-based scheme, border-aware scheme for different network sizes.

Broadcast parameters for different schemes are selected according to experiment. It is ensured first that the broadcast reach ability and data delivery ratio are larger than 90%. In the situation, when counter threshold is 3, counter-based scheme achieves its best performance in terms of other metrics. In this way, distance threshold is selected as 20m for distance-based scheme; counter threshold and distance threshold are selected as 2 and 20 for border-aware scheme. For energy-aware scheme, select the same counter and distance thresholds as border-aware scheme, β as 0.8 and α as 0.2.

Fig. 2 and Fig. 3 show the performance comparison of DD with different broadcast schemes. This scheme has an average dissipated energy and average end-to-end delay of data packet compared with border-aware. Border-aware and energy-aware schemes substantially reduce the energy consumption

due to more saved redundant broadcasts. Because they select the further nodes to rebroadcast, the number of hops from the source to the sink is less than other schemes resulting in less average end-to-end delay of data packet. Although flooding can find the lowest delay path, it has poor average end-to-end delay of data packet due to more contention and collision when exploratory data are sent.

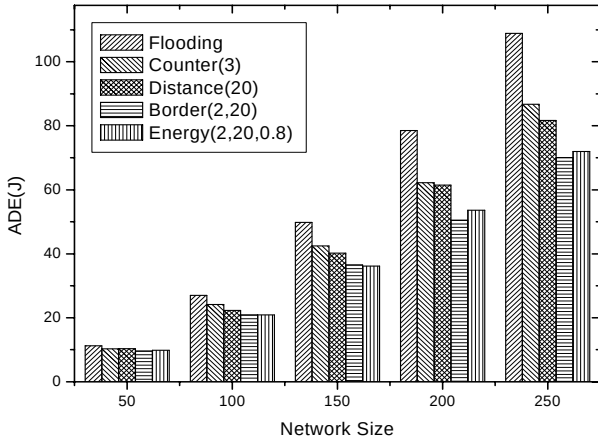


Fig. 2 Average dissipated energy (network size)

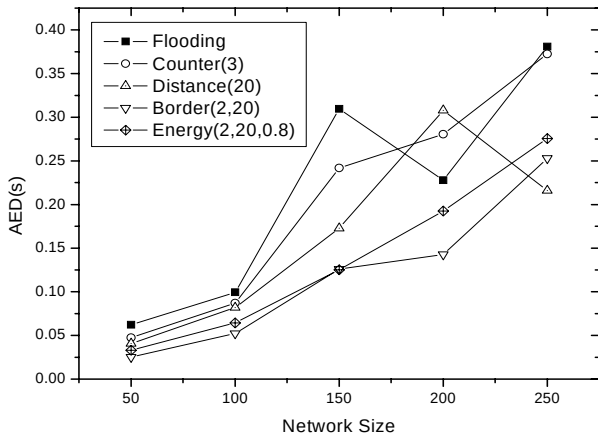


Fig. 3 Average end-to-end delay of data packet (network size)

Fig. 4 and Fig. 5 show the performance comparison of different broadcast schemes as a function of network density. The experiment exhibits similar results with the first experiment.

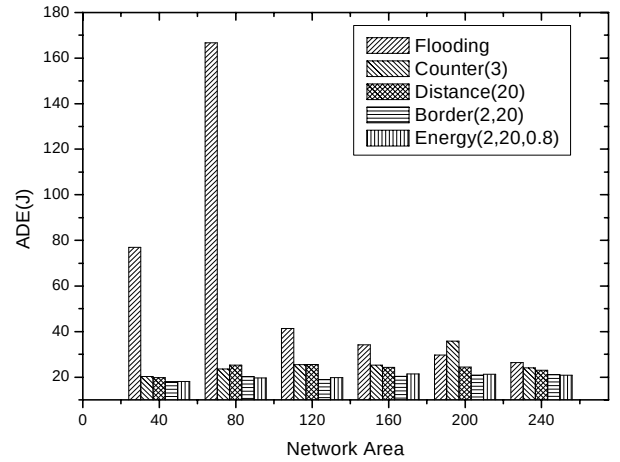


Fig. 4 Average dissipated energy (network density)

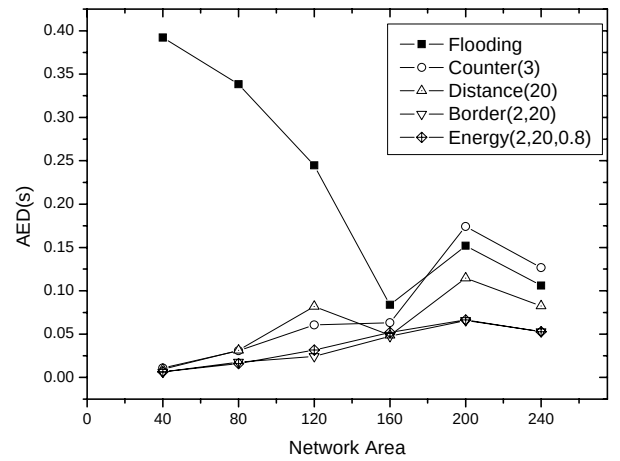


Fig. 5 Average end-to-end delay of data packet (network density)

The above simulation results show that energy-aware and border-aware broadcast schemes have high efficiency, scalability and adaptability. But border-aware has poor energy-balancing. As the traffic load increases, border-aware scheme speeds up the death of nodes which are further from the broadcast transmitter. After WSN works for a certain time, the nodes will have different energy due to different work load or new added nodes. Border-aware scheme could not balance the energy consumption among the nodes resulting in shorter network lifetime.

In the third experiment, the network lifetime of DD is compared with the two interest broadcast schemes. Select 100 nodes 240×240m sensor field in the second

experiment as the experimental scenario. The initial energy of nodes is classified into two levels: 20J and 10J, the sink and the source nodes have 20J, other nodes have 10J. Fig. 6 shows the impact of traffic load on network lifetime. Traffic load increases with the reducing of event interval. Energy-aware broadcast scheme has longer network lifetime than border-aware in different traffic loads. But the improvement on network lifetime is not outstanding due to the fact energy consumption of nodes for overhearing and idle is the main part of energy consumption of nodes. If appropriate sleep schedule mechanism is used for DD, this scheme will perform better.

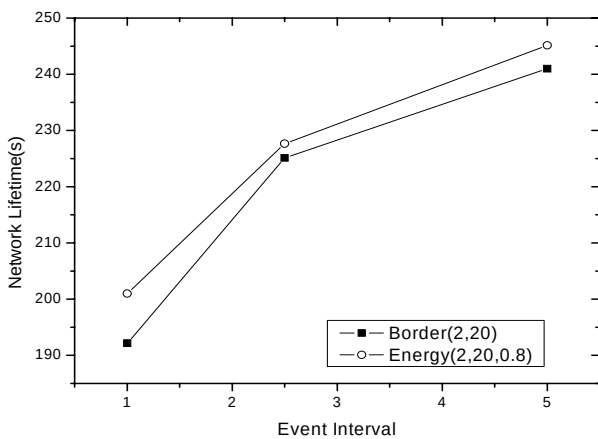


Fig. 6 Network lifetime

5. Conclusions and Future Work

This paper compares the performances of several broadcast schemes combining with directed diffusion and proposes an energy-aware broadcast scheme adapting to directed diffusion. This scheme combines the advantages of distance-based and counter-based broadcast schemes, and considers the remainder energy of sensor node. Inspired by directional backoff concept in border-aware broadcast scheme, segment delay is proposed which is associated with distance and remainder energy.

Simulation results show that the proposed scheme has the following advantages: (1) simple, efficient and scalable, (2) more energy-balancing than other

schemes, (3) longer network lifetime than other schemes.

In the future, sleep schedule policy combining with DD will be considered.

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